

# CSP-TRUST-CONCEPT

Execution Attestation and Distributed Trust Layer for Cognitive Skills

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## Abstract

CSP-0.1 and CSP-0.2 define portable and verifiable cognitive skills through immutable Passports, Runtime behavior, and Compatibility Layers. However, portability alone does not establish trust. This document proposes an optional **Execution Attestation Layer (EAL)** — a distributed trust mechanism that allows runtime environments to issue verifiable attestations about skill behavior without modifying the original Passport. This proposal does not define a finalized trust protocol and is intended for ecosystem discussion and future experimentation.

## 1 Introduction

A receiving system may understand a skill structurally while still lacking confidence regarding runtime stability, security, behavioral consistency, interoperability history, and prior successful execution in other environments.

The purpose of the Execution Attestation Layer is to enable runtime ecosystems to issue verifiable execution attestations for cognitive skills.

**This document is exploratory and does not constitute an official CSP protocol version.**

## 2 Core Principle

The fundamental principle is that the **Passport remains immutable**. Attestations are external ecosystem observations.

| Layer         | Purpose                     | Owner             |
|---------------|-----------------------------|-------------------|
| Passport      | Identity and requirements   | Author            |
| Compatibility | I/O and operational support | Author            |
| Runtime       | Dynamic execution metrics   | Local environment |
| Attestation   | Historical trust evidence   | Ecosystem         |

## 3 Execution Attestation

An attestation is a signed statement produced by a runtime environment confirming successful execution, validation, or observation of a skill under specific conditions.

```

1  attestation:
2    issuer: "Gemini"
3    type: "runtime_verified"
4    passport_hash: "sha256:..."
5    runtime_tdi: 0.12
6    compatibility_verified: true
7    malicious_behavior_detected: false
8    date: "2026-05-08"

```

Listing 1: Example Attestation Structure

The attestation:

- does **NOT** modify the Passport,
- does **NOT** redefine the skill,
- acts only as external trust metadata.

## 4 Motivation and Design Constraints

Modern AI ecosystems suffer from several trust-related problems, relying primarily on reputation, stars, downloads, or informal testing. This proposal explores a more structured trust mechanism.

The proposal intentionally avoids modifying Passport immutability, introducing centralized trust authorities, forcing mandatory attestation infrastructure, or defining cryptographic standards prematurely. It is an exploratory architectural proposal intended to document the idea and stimulate ecosystem discussion.

## 5 Discussion and Conclusion

Portable cognitive skills require more than structural compatibility — they require observable execution trust. The Execution Attestation Layer explores the possibility of building a distributed trust network for interoperable cognitive systems.

## Related Works

- **CSP-0.1** — DOI: [10.5281/zenodo.19639060](https://doi.org/10.5281/zenodo.19639060)
- **CSP-0.2** — DOI: [10.5281/zenodo.20034069](https://doi.org/10.5281/zenodo.20034069)

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Methodological framework: **Method of Artsybashev (AAM-V1)**.

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